

Informe de experimento: results_validation_gpt-5.4_stress

Generado el 2026-09-15 23:42 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_stress (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
low-agent/benchmarks/results_validation_gpt-5.4_stress
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate, pfagent
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=['stress'] (1 report)
- Items por dificultad: stress=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-15 23:40
- Costo total: 0.5639 USD (10 items; costo medio 0.05639 USD/item)
- Items totales: 10; ok=10; con error=0; tokens totales=199722
- Solved global: 60.0% (6/10)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) –1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text –90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del scoreboard_per_case), en los 4 grupos de metricas_pfagent_definicion utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); corrección solver-grounded Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); fidelidad Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); costo Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	\$
react_nogate	case14	5	60.0 (3/5)	–	0.01	60.0 (3/5)	100.0 (5/5)	4.87e-06	96.7	100.0 (5/5)	3.4	13485	0.0 (0/5)	0.0 (0/3)	0.1925
pfagent	case14	5	60.0 (3/5)	–	0.01	60.0 (3/5)	100.0 (5/5)	4.87e-06	–	100.0 (5/5)	4.0	26460	0.0 (0/5)	100.0 (3/3)	0.3714

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	stress
react_nogate	60.0 (3/5)
pfagent	60.0 (3/5)

Solved por dificultad (% , items resueltos / items)

Method	stress
react_nogate	60.0 (3/5)
pfagent	60.0 (3/5)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

pass_first = pasó al primer intento; **pass_retry** = pasó tras el único reintento con el veredicto añadido; **abstained** = falló dos veces, respuesta forzada a fallo declarado sin números. N = items con **final_gate** activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained
pfagent	5	0.0 (0/5)	0.0 (0/5)	100.0 (5/5)

Tipos de error de formulación (conteo de items)

Method	extra_arg	changes_result	ok
react_nogate	2		3
pfagent	2		3

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↳ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5624 chars más]
  gate: FAIL (reasons=isolated_or_undefined_buses, mismatch=9.78e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded the IEEE 14-bus system and disconnected the branch between bus 7 and bus 8; the power flow converged, but bus 8 became
↳ electrically isolated.

2. Key numbers
- Total load: 252.656 MW
- Total generation: 265.463 MW
- Total losses: 12.807 MW
- Voltage range: ...[608 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded the IEEE 14-bus system and disconnected the branch between bus 7 and bus 8; the power flow converged, but bus 8 became
↳ electrically isolated.

2. Key numbers
- Total load: 252.656 MW
- Total generation: 265.463 MW
- Total losses: 12.807 MW
- Voltage range: valid solved bus voltages span 1.010 pu to 1.070 pu
- Special note: bus 8 voltage is undefined (NaN) after the disconnection, indicating isolation

3. Violations, if any
- Voltage violations:
  - Bus 1: 1.060 pu, overvoltage
  - Bus 6: 1.070 pu, overvoltage
  - Bus 11: 1.0503 pu, overvoltage
  - Bus 12: 1.0545 pu, overvoltage
- No thermal branch violations were reported
```

- Disconnected branch 7-8 now carries 0.0 MW / 0.0 MVar

4. Suggested next step

A useful next step is to reconnect branch ...[108 chars más]

--- Scoring ---

formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=100.0 %; solved=True (declared_failure); flags=-; error=-; tokens=5250;
→ cost_usd=0.0171; wall=7.4 s

Ejemplo: formulación fallida (formulation_exact=False)

request (case14-stress-002-s0, case14, difficulty=stress, seed=1627694679): Load the 14-bus test case (case14), set the active load at bus
→ 9 to 37.7 MW, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line whose
→ loading is above 80%.

intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7}) -> disconnect_line({"from_bus": 7, "to_bus":
→ 8}) -> run_powerflow({})

executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7, "q_mvar": 16.6}) ->
→ disconnect_line({"from_bus": 7, "to_bus": 8}) -> run_powerflow({})

fuentes de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
→ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)

--- Ronda 2 ---

tool: modify_load({"bus_id": 9, "p_mw": 37.7, "q_mvar": 16.6})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.209724513341782, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_p ...[6267 chars más]
gate: PASS (reasons=-, mismatch=7.05e-09 MW)

--- Ronda 3 ---

tool: disconnect_line({"from_bus": 7, "to_bus": 8})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.218146717892699, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_p ...[5516 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.57e-08 MW)

--- Ronda 4 ---

tool: run_powerflow({})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.218146717892699, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_p ...[5516 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.57e-08 MW)

--- Ronda 5 ---

LLM: 1. Summary: I loaded case14, set bus 9 active load to 37.7 MW using the existing reactive load at that bus, disconnected branch 7-8,
→ reran the power flow successfully, and found no connected line loaded above 80%.

2. Key numbers:

- Total load: 264.4278 MW
- Total generation: 278.6065 MW
- Total los ...[500 chars más]
--- Respuesta final ---

1. Summary: I loaded case14, set bus 9 active load to 37.7 MW using the existing reactive load at that bus, disconnected branch 7-8, reran
→ the power flow successfully, and found no connected line loaded above 80%.

2. Key numbers:

- Total load: 264.4278 MW
- Total generation: 278.6065 MW
- Total losses: 14.1787 MW
- Voltage range: 1.0102 to 1.0700 pu among solved buses
- Note: bus 8 is isolated after disconnecting 7-8, so its voltage is reported as NaN and excluded from the range

3. Violations:

- Lines above 80% loading: none
- Voltage violations:
- Bus 1: 1.0600 pu overvoltage
- Bus 2: 1.0700 pu overvoltage
- Bus 12: 1.0545 pu overvoltage

4. Suggested next step:

- If you want, I can list all line loadings in descending order or check the most heavily loaded line after this outage.

--- Scoring ---

formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=0.03; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
→ tokens=20047; cost_usd=0.0544; wall=8.5 s

Ejemplo: con bandera de fallo (claimed/stale/abstained)

request (case14-stress-003-s0, case14, difficulty=stress, seed=1911784258): Load the 14-bus test case (case14), disconnect the line between
→ bus 1 and bus 2, increase the load at bus 3 to 282.6 MW and 57 Mvar (three times its nominal value), and finally rerun the power flow,
→ then report the bus with the lowest voltage magnitude.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
→ 282.6, "q_mvar": 57.0}) -> run_powerflow({})

executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
→ 282.6, "q_mvar": 57.0}) -> run_powerflow({})

fuentes de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
→ "total_gen_capacity_mw": 772.4000000000001}

```

gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -34.6873802729158, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5949 chars más]
gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 3, "p_mw": 282.6, "q_mvar": 57})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↪ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
  ↪ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 4 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↪ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
  ↪ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 5 ---
LLM: 1. **Summary:** I loaded **case14**, disconnected **line 1-2**, increased **bus 3** load to **282.6 MW / 57 Mvar**, and the final AC
  ↪ power flow **did not converge**, so the lowest-voltage bus cannot be determined for that modified state.

2. **Key numbers:**
- After loading the base case, total lo ...[1066 chars más]
--- Respuesta final ---
1. **Summary:** I loaded **case14**, disconnected **line 1-2**, increased **bus 3** load to **282.6 MW / 57 Mvar**, and the final AC power
  ↪ flow **did not converge**, so the lowest-voltage bus cannot be determined for that modified state.

2. **Key numbers:**
- After loading the base case, total load was **253.8458 MW**.
- After disconnecting line **1-2** but before the load increase, the solved case had:
  - **Total load:** **253.8458 MW**
  - **Total generation:** **292.0857 MW**
  - **Total losses:** **38.2400 MW**
  - **Voltage range:** **0.9955 pu** at bus **5** to **1.0900 pu** at bus **8**
- After increasing bus 3 load to **282.6 MW / 57 Mvar** and rerunning, the solver reported:
  - **Power flow failed: LoadflowNotConverged: Power Flow nr did not converge after 10 it ...[566 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=-; F_MAE=-; faithful=100.0 %; solved=True (declared_failure); flags=stale_state; error=-;
  ↪ tokens=13138; cost_usd=0.0388; wall=11.1 s

```

Método pfagent

Ejemplo: resuelto (solved=True)

```

request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
  ↪ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
  ↪ "total_gen_capacity_mw": 772.40000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va": ...[5624 chars más]
gate: FAIL (reasons=isolated_or_undefined_buses, mismatch=9.78e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
The IEEE 14-bus system was loaded, and the branch between bus 7 and bus 8 was disconnected; the power flow converged, but bus 8 became
  ↪ electrically isolated with undefined voltage.

2. Key numbers
- Total load: 252.656 MW
- Total generation: 265.463 MW
- Total losses: ...[653 chars más]
--- Ronda 3 ---
LLM: 1. One-sentence summary
The IEEE 14-bus system was loaded and the 7-8 transformer branch was disconnected; although the solver returned a converged state, the
  ↪ result is not fully valid for acceptance because bus 8 is isolated from the slack and has no valid voltage.

2. Key numbers
- Total load: ...[760 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
  ↪ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=-; F_MAE=0.00; faithful=- %; solved=True (declared_failure); flags=abstained,abstained_on_solvable;
  ↪ error=-; tokens=9209; cost_usd=0.0308; wall=9.6 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-stress-002-s0, case14, difficulty=stress, seed=1627694679): Load the 14-bus test case (case14), set the active load at bus
↳ 9 to 37.7 MW, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line whose
↳ loading is above 80%.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7}) -> disconnect_line({"from_bus": 7, "to_bus":
↳ 8}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 9, "p_mw": 37.7, "q_mvar": 16.6}) ->
↳ disconnect_line({"from_bus": 7, "to_bus": 8}) -> run_powerflow({}) -> get_status({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 9, "p_mw": 37.7, "q_mvar": 16.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.209724513341782, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6267 chars más]
gate: PASS (reasons=, mismatch=7.05e-09 MW)
--- Ronda 3 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.218146717892699, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5516 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.57e-08 MW)
--- Ronda 4 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.218146717892699, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5516 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.57e-08 MW)
--- Ronda 5 ---
LLM: 1. One-sentence summary

I loaded case14, set bus 9 active load to 37.7 MW, disconnected branch 7-8, reran the power flow successfully, and found no in-service line
↳ loaded above 80%.

2. Key numbers

- Total load: 264.42776694368655 MW
- Total generation: 278.6065135958651 MW
- Total losses: 14.1787 ...[734 chars más]
--- Ronda 5 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
↳ "total_load_mw": 264.42776694368655, "total_gen_capacity_mw": 772.4000000000001, "has_last_result": true, "last_result_summary":
↳ {"converged": true, "total_load_mw": 264 ...[182 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 6 ---
LLM: 1. One-sentence summary

I loaded case14, set bus 9 active load to 37.7 MW, disconnected branch 7-8, and the resulting power flow is not fully valid for the whole
↳ 14-bus network because bus 8 is isolated and has no valid voltage, so no accepted final branch-loading report can be given for the full
↳ s ...[740 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
↳ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=0.03; faithful=-- %; solved=False (formulation);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=38158; cost_usd=0.1042; wall=15.7 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-stress-001-s0, case14, difficulty=stress, seed=827308000): Load the 14-bus test case (case14), disconnect the line between
↳ bus 2 and bus 4, disconnect the transformer branch between bus 7 and bus 8, and finally rerun the power flow, then list every line
↳ whose loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 2, "to_bus": 4}) -> disconnect_line({"from_bus": 7,
↳ "to_bus": 8}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 270.1361769951196,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 2, "to_bus": 4})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.717168458565059, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5946 chars más]
gate: PASS (reasons=, mismatch=3.08e-08 MW)
--- Ronda 3 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.733791919883227, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5459 chars más]

```

```

gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=8.78e-08 MW)
--- Ronda 4 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.733791919883227, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_p ...[5459 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=8.78e-08 MW)
--- Ronda 5 ---
LLM: 1. One-sentence summary

I loaded IEEE case14, disconnected branch 2-4 and transformer branch 7-8, reran the power flow, and found two lines with loading above 100%.

2. Key numbers

- Total load: 270.1361769951196 MW
- Total generation: 287.41789199699963 MW
- Total losses: 17.281715089657098 MW
- ...[740 chars más]
--- Ronda 5 ---
LLM: 1. One-sentence summary

I loaded IEEE case14, disconnected branch 2-4 and branch 7-8, reran the power flow, and the solved network shows two branches with loading
  ↪ above 100%, while bus 8 has no valid solved voltage because it is isolated after the 7-8 disconnection.

2. Key numbers

- Total load: 2 ...[943 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
  ↪ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=True (declared_failure); flags=abstained,abstained_on_solvable;
  ↪ error=--; tokens=28712; cost_usd=0.0821; wall=13.4 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- El gate verificó 26 resultados de flujo, 15 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved más alto: **pfagent** 60.0% (3/5); más bajo: **react_nogate** 60.0% (3/5).
- Formulación exacta agregada sobre métodos con tools: 60.0% (6/10).
- Errores de formulación más frecuentes: **extra_arg_changes_result**=4.
- Ningún item reclamó éxito sobre un fallo (**claimed_success_on_failure**=0).
- De 10 items en estado de fallo, 10 lo declararon de forma segura (SFR 100.0% (10/10)).
- 2 de 10 items con mutación de red reportaron estado obsoleto (**stale_state**).
- Trazabilidad de números (Faith.) media con tools: 96.7%.
- 10 items esperaban un resultado no convergido (**islanded**=6, **non_converged**=4); el método lo declaró correctamente (**solved**) en 60.0% (6/10), con **solved_reason declared_failure**=6.
- Costo total 0.5639 USD; el método más caro fue **pfagent** con 0.3714 USD (66%).
- Métodos con tools: 3.70 tool calls medias por item (máximo 5).

6. Archivos

Rutas absolutas bajo **results_validation_gpt-5.4_stress.report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ /power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard_per_case.json
traces (10 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
  ↪ power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/traces

```